

Executive Summary: Exploratory Data Analysis

Track: Data Science | Program: EncoderX Batch 02 | Dataset: US Sample Superstore

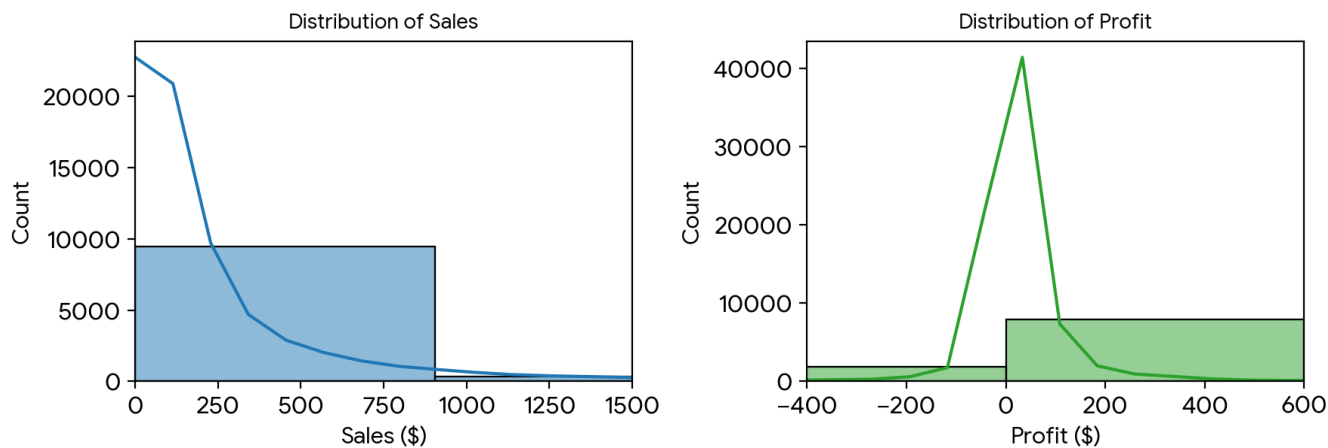
1. Project Overview & Data Preprocessing

This report presents an in-depth Exploratory Data Analysis (EDA) conducted on the retail transactions of a US Superstore. The dataset consists of **9,994 transactions** spanning multiple product categories, customer segments, and geographic regions. Data cleaning operations removed 17 duplicate records, verified zero missing entries, and pruned non-analytical columns (Country, Postal Code) leaving a clean dataset of **9,977 verified rows**.

Metric	Value	Key Observation
Total Transactions	9,977	Cleaned rows across 4 regions
Total Revenue	\$2,296,195.59	Highest in West and East regions
Total Profit	\$286,241.42	Overall profit margin ~12.46%
Average Discount	15.63%	Discounts > 20% severely impair profitability

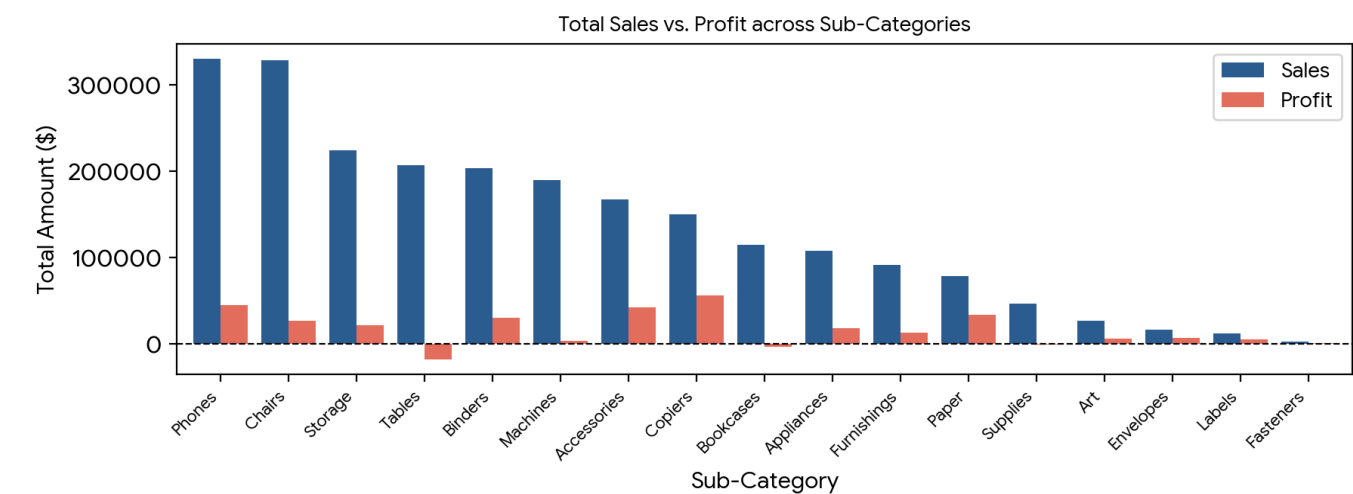
2. Visual Analytics & Diagnostic Findings

Figure 1: Distribution Analysis of Sales & Profit



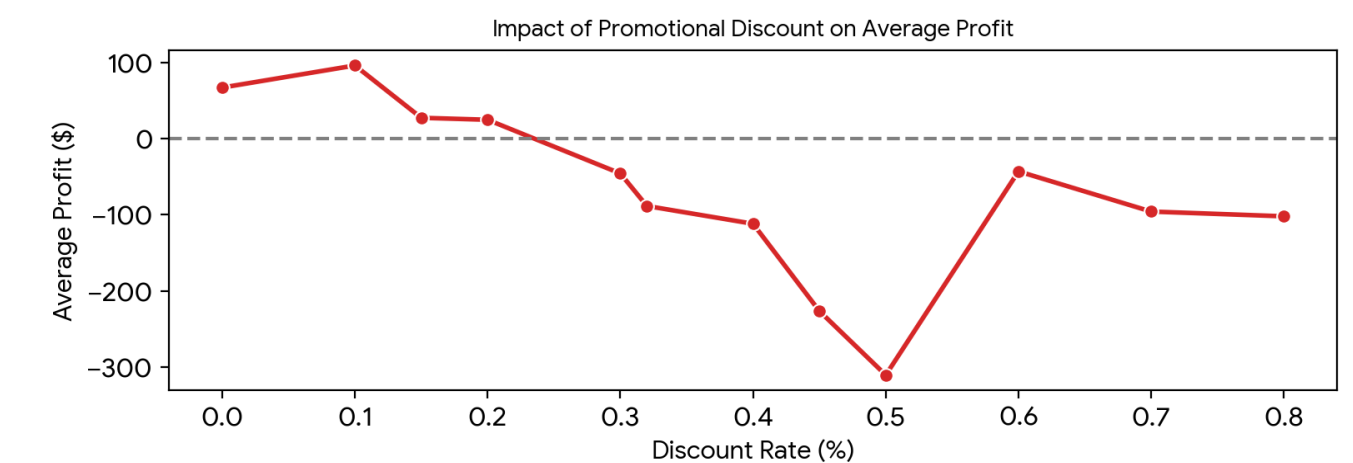
- *Right-Skewed Distributions:* Both Sales and Profit are heavily positively skewed. High-frequency low-ticket orders form the bulk of transactions, while rare enterprise orders drive substantial profit spikes.

Figure 2: Performance Across Sub-Categories (Sales vs Profit)



- *Category Inefficiencies:* **Phones** and **Chairs** lead in revenue generation. However, sub-categories like **Tables** (\$-17.7k profit) and **Bookcases** (\$-3.4k profit) suffer consistent net losses despite solid sales.

Figure 3: Profitability Decay vs Discount Rate



- *The 20% Inflexion Threshold:* As discounts cross beyond 20%, average order profitability plunges into negative territory, confirming that deep discounting erodes profit without proportionally driving sustainable volume.

3. Strategic Action Plan & Recommendations

1. **Implement Discount Governance:** Cap standard sales representative discounts at **15%–20%**. Require regional manager sign-off for enterprise discounts above 20% to curb negative-margin sales.
2. **Restructure Loss-Making Categories:** Revisit shipping and supplier contracts for **Tables** and **Bookcases**. Bundle low-margin furniture with high-margin items (e.g., Office Accessories/Copiers) to protect bottom-line margins.
3. **Prioritize High-Yield Regions:** Direct marketing investment into the **West** and **East** regions, which demonstrate the highest customer lifetime value and profit resilience.